Yianni Karabatis

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Education

Ph.D. in Computer Science — Aug 2022 – May 2028 (Planned)
University of Maryland — *College Park, MD*
- Advised by Yiannis Aloimonos
- Research Interests: Multimodal Vision & Audio

M.S. in Computer Science — Aug 2022 - Dec 2024
University of Maryland — *College Park, MD*
- GPA: 3.77

B.S. in Computer Science — Aug 2017 - May 2021
University of Maryland, Baltimore County — *Baltimore, MD*
- GPA: 3.74

Current Research

Event-based Visual Microphone — 2025–2026
First-author research
- Reconstructed intelligible speech from only an event camera (μs resolution) observing laser speckle vibrations on a chip bag, without an acoustic sensor. Achieved STOI 0.83 vs. 0.67 from other works.
- Built the first audio/event-based vision dataset and engineered μs-scale synchronization to 0.316 μs RMS to the played waveform.
- Characterized the physical recoverable spectral shape of the chip bag.
- Model: a learned per-pixel 25 ms filterbank (1D conv over time) with a spatial CNN and a bidirectional transformer decoder (17.6 M parameters).

Peer-Reviewed Publications

Karabatis, Y., Burner, L., Dessalene, E., Nolan, M. Metzler, C., Fermuller C., Duraiswami, R. & Aloimonos, Y. (2027). EvSpeech: Learning to Reconstruct Speech with an Event Camera. Submitted to *ICASSP 2027*.

Dessalene, E., He, B., Maynord, M., Tussa, Y., Matripaganda, P., **Karabatis, Y.**, Roy, N., & Aloimonos, Y. (2026, March). FEEL (Force-Enhanced Egocentric Learning): A Dataset for Physical Action Understanding. *In 2026 European Conference on Computer Vision (ECCV).* IEEE.

Karabatis, Y., Lin, X., Sanket, N. J., Lagoudakis, M. G., & Aloimonos, Y. (2023, October). Detecting Olives with Synthetic or Real Data? Olive the Above. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 4242–4249). IEEE.

Karabatis, Y.*, Lin, X.*, & Aloimonos, Y. (2023, October). Agrifood Detection Enhancement with Synthetic Data. In *2023 Present and Future of Agricultural Robotics and Technologies: Workshop of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE.

Work Experience

Machine Learning Intern — Jun 2024 – Aug 2024
Center for Advanced Transportation Technology — *College Park, MD*
- Designed, trained, and tested machine learning models to perform anomaly detection on highway incidents using dash-cam footage.
- Models achieved 95% overall accuracy, resulting in faster incident reporting on Maryland roads.

Fulbright Research Fellow — Aug 2021 – Jul 2022
Fulbright Foundation — *Technical University of Crete, Greece*
- Generated and hand-labeled a dataset of thousands of olives from actual olive groves.
- Created the world's first model to segment olives on actual olive trees autonomously.
- Implemented a pipeline to effortlessly generate thousands of auto-labeled synthetic images using Blender and image-to-image translation.
- Created state-of-the-art UNet models to semantically segment olives on real-life olive trees, resulting in strong performance.

Undergraduate Research Assistant — May 2020 – Aug 2020
National Science Foundation - Research Experience for Undergraduates — *DIMACS, Rutgers University*
- Created an impedance controller using machine learning models for the Baxter robot to smooth the motion of an arm under resistance. The result was an enhanced robot arm response under force.

Software Engineering Intern — Jun 2018 – Aug 2018
Northrop Grumman — *Woodlawn, MD*
- Used MS SharePoint and InfoPath to design and implement a user interface to collect data to populate a back-end database for internal operations.

Intern — May 2017
Vheda Health — *Columbia, MD*
- Researched several inventory management systems, innovative technologies, and case studies for stabilizing high-risk pregnancies with telemedicine.

Skills

Programming Languages and Libraries

Python, C++, PyTorch, Tensorflow, OpenCV, Scikit-Learn, Numpy

Software

Prophesee Metavision for Event Cameras, Blender, GIS (multispectral bands)

Research

- Computer Vision, Artificial Intelligence, Deep learning, Machine Learning, Audio Processing, Event-based Vision
- Video-to-Audio Generation, Image-to-Image Translation, Generative Adversarial Networks, Synthetic Data Generation, Segmentation

Relevant Coursework

University of Maryland

- Multimodal Computer Vision
- Classical Computer Vision
- Modern Computer Vision
- Numerical Optimization
- Advances in Extended Reality (XR)
- Machine Learning Paradigms
- Computational Imaging
- Computational Audition

Awards and Honors

Fulbright Research Scholar Aug 2022 - Jul 2023

Fulbright Foundation

Dean's Fellowship Aug 2022

University of Maryland

Chair's Fellowship Aug 2022

University of Maryland

Louis Stokes Alliance for Minority Participants (LSAMP) Fellowship May - Aug 2019

University System of Maryland

President's List 2018, 2019, 2020, 2021

University of Maryland, Baltimore County

Merit Scholar Aug 2017 - May 2021

University of Maryland, Baltimore County

Selected Projects

Lensless Facial Recognition 2024

Course Project

- Facial recognition using images captured without a camera lens (very noisy) for privacy preservation.
- Converted lensless images to the frequency domain via Discrete Cosine Transform (DCT).
- Trained and tested Vision Transformers to perform facial recognition on the DCT, achieving 90% accuracy.

GTA to Real Life Driving 2024

Course Project

- Make images from GTA5 look more realistic.
- Trained numerous GANs on the GTA5 dataset to improve nature scene lighting, object colors, car glossiness, and road texture while minimizing hallucinations.

Gabor Wavelet Image-to-Image Translation 2023

Course Project

- Created loss functions for the task of image generation to preserve Gabor wavelets in the output. This resulted in decreased visual hallucinations on natural datasets.

Breast Cancer Classification 2019

University of Maryland, Baltimore County

- Design Convolutional Neural Networks to classify H&E-stained histopathological images from Kaggle.
- Results of 91% accuracy.
- Presented at UMBC's Undergraduate Research and Creative Achievement Day (URCAD) and featured in the *UMBC Magazine*

Teaching Assistantships

Computer Vision Aug 2024 – Dec 2024
CMSC426 *University of Maryland*
- Giving lectures
- Holding weekly office hours
- Grading projects and homework

Advanced Data Structures Aug 2022 – May 2023
CMSC420 *University of Maryland*
- Holding weekly office hours
- Grading projects and homework

Introduction to Web Development Feb 2025 – present
CMSC335 *University of Maryland*
- Holding weekly office hours
- Grading projects and homework

Bits and Bytes of Computer Science Aug 2023 – Dec 2023
CMSC100 *University of Maryland*
- Giving guest lectures.

Introduction to Computer Systems Feb 2024 – May 2024
CMSC216 *University of Maryland*
- Leading discussion sections.
- Holding weekly office hours.
- Grading projects and homework.

Research Initiatives

Student Participant at the NSF NeuroAI Workshop 2025
- Participated in the Embodied Intelligence discussion groups.
- Prepared and presented a SWOT analysis on the field of Neuroscience-based Artificial Intelligence from a student perspective.
- Contributed a personal statement to the workshop's final report.

Reviewer for IEEE Robotics and Automation Letters (RA-L) 2025
Reviewer for IEEE Transactions in Image Processing (TIP) 2026

Volunteering

Brook Codes Hackathon 2023
- Led a session called "Introduction to Research" for high school students interested in research

Richard Tapia Celebration of Diversity in Computing Conference 2022
Student Volunteer

CSPS Grand Lodge 2019
Spring clean-up of the cemetery

First Lego League (FLL) 2018
Scorekeeper for a youth robotics competition

Extracurriculars

Affiliate of Meyerhoff Scholars Program 2017 – 2021
Byzantine Choir Chanter 2016 – present
Violinist 2008 – present